

Notes: Gao & Huang (RFS, 2020)

Informing the Market: The Effect of Modern Information Technologies on Information Production

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1 Big picture

- **Question.** Do modern information technologies merely disseminate existing information, or do they also change incentives for outsiders to produce information?
- **Shock.** The paper uses the staggered implementation of the SEC's EDGAR electronic filing system from 1993 to 1996 as a shock to public dissemination of corporate disclosures.
- **Economic tension.** Easier access to filings can crowd out private information production because public information becomes widely known. But it can also crowd in production by lowering search and processing costs.
- **Main finding.** EDGAR increases information production by corporate outsiders rather than crowding it out.
- **Individual investors.** Individual net buying after earnings announcements becomes more informative about future abnormal returns after a firm becomes an EDGAR filer.
- **Who drives it?** The effect is stronger for investors with internet access and for investors with lower reliance on raw public information, consistent with actual information production rather than pure attention or naive public-information trading.
- **Analysts.** Analyst coverage and forecast accuracy rise after EDGAR, and the market response to analyst forecast revisions becomes stronger.
- **Robustness.** Results survive propensity-score matching, group-specific trends, falsification tests, alternative EDGAR timing definitions, and excluding transitional filers.
- **Market efficiency.** EDGAR reduces price synchronicity, absolute return autocorrelation, and pricing errors, suggesting improved price efficiency.
- **Takeaway.** Digital dissemination of mandatory disclosures can increase both the amount and quality of information produced by market participants.

2 Data and measurement

2.1 EDGAR implementation

- Before EDGAR, corporate filings were paper documents submitted to the SEC, with access concentrated in SEC public reference rooms.
- The SEC introduced EDGAR in 1993 and phased in mandatory electronic filing in ten groups from April 1993 to May 1996.
- The treatment variable $\text{PostEDGAR}_{i,t}$ equals one after firm i becomes a mandatory EDGAR filer.
- The staggered phase-in creates treated and not-yet-treated firms at the same calendar dates.

Interpretation: The design turns a regulatory technology rollout into a difference-in-differences setting. The key identifying claim is that firms in later EDGAR cohorts provide a counterfactual for firms already switched to electronic filing.

2.2 Individual-investor trading sample

- The individual trading sample contains household brokerage trades around earnings announcements.
- The main trading variable is $\text{Netbuy}_{[+1,+20]}$: shares bought minus shares sold by individual investors during days +1 to +20 after an earnings announcement, scaled by shares outstanding.

- Future stock-price informativeness is measured using DGTW characteristics-adjusted cumulative returns starting on day +21.
- The paper uses four return horizons: 3 months, 6 months, 12 months, and 18 months.

Interpretation: If individual trades become more informative after EDGAR, the relation between post-announcement net buying and later abnormal returns should strengthen after the firm becomes an EDGAR filer.

2.3 Internet users and information producers

- Internet users are investors who have previously placed trades through the internet.
- Nonusers are investors without a history of internet trading.
- The paper also splits investors into high-RPI and low-RPI investors.
- RPI is a reliance-on-public-information measure adapted from Kacperczyk and Seru: high-RPI investors' trades line up more with analyst recommendation changes, while low-RPI investors rely less on raw public signals.

Interpretation: If EDGAR lowers information-processing costs, the impact should be stronger for internet users and for investors more likely to process information rather than simply mimic public signals.

2.4 Analyst research sample

- Analyst data come from I/B/E/S quarterly earnings forecasts.
- The paper studies three analyst information-production outcomes:
 - number of analysts covering a firm;
 - forecast accuracy, defined as the negative absolute forecast error scaled by stock price;
 - market response to analyst forecast revisions, measured by DGTW-adjusted returns around revision events.

Interpretation: Analyst coverage captures the quantity of analyst production; forecast accuracy and revision returns capture quality and informativeness.

3 Models and empirical specifications

3.1 Individual-investor informativeness

The baseline individual-investor specification is:

$$CAR_{i,t+h} = \alpha_i + \lambda_t + \beta_1 \left(\text{Netbuy}_{i,t,[+1,+20]} \times \text{PostEDGAR}_{i,t} \right) + \beta_2 \text{Netbuy}_{i,t,[+1,+20]} + \beta_3 \text{PostEDGAR}_{i,t} + \gamma' X_{i,t} + \varepsilon_{i,t}. \quad (1)$$

- $CAR_{i,t+h}$ is future DGTW-adjusted return over a 3-, 6-, 12-, or 18-month horizon.
- α_i are firm fixed effects.
- λ_t are year-quarter fixed effects.
- $X_{i,t}$ includes firm characteristics and earnings-surprise controls.

Interpretation: The coefficient β_1 is the main information-production coefficient. If EDGAR makes individual trades more informative, $\beta_1 > 0$.

3.2 Heterogeneity by internet access

The paper replaces aggregate Netbuy with net buying by internet users and nonusers:

$$CAR_{i,t+h} = \alpha_i + \lambda_t + \beta_I(\text{Netbuy}_{i,t}^{\text{Internet}} \times \text{PostEDGAR}_{i,t}) + \beta_N(\text{Netbuy}_{i,t}^{\text{Nonuser}} \times \text{PostEDGAR}_{i,t}) + \dots \quad (2)$$

Interpretation: The direct technology channel predicts $\beta_I > \beta_N$, because internet users benefit most directly from electronic filing access.

3.3 Heterogeneity by reliance on public information

The high-RPI/low-RPI specification is analogous:

$$CAR_{i,t+h} = \alpha_i + \lambda_t + \beta_H(\text{Netbuy}_{i,t}^{\text{HighRPI}} \times \text{PostEDGAR}_{i,t}) + \beta_L(\text{Netbuy}_{i,t}^{\text{LowRPI}} \times \text{PostEDGAR}_{i,t}) + \dots \quad (3)$$

Interpretation: If EDGAR increases true information production, the effect should be stronger for low-RPI investors, who are less likely to simply follow public analyst signals.

3.4 Analyst research quantity and quality

For analyst coverage and forecast accuracy, the paper estimates:

$$Y_{i,t} = \alpha_i + \lambda_t + \beta \text{PostEDGAR}_{i,t} + \gamma' X_{i,t} + u_{i,t}, \quad (4)$$

where $Y_{i,t}$ is either number of analysts or forecast accuracy.

Interpretation: A positive β means that analysts produce more research or more accurate forecasts after a firm's filings become electronically available.

3.5 Market response to analyst revisions

For revision informativeness, the paper estimates:

$$CAR_{i,a,t[-1,+1]}^{\text{revision}} = \alpha_i + \delta_{a,t} + \beta(\text{Revision}_{i,a,t} \times \text{PostEDGAR}_{i,t}) + \text{hetaRevision}_{i,a,t} + \gamma' X_{i,t} + e_{i,a,t}. \quad (5)$$

Interpretation: A positive coefficient on $\text{Revision} \times \text{PostEDGAR}$ means analyst revisions become more informative after EDGAR.

3.6 Pricing efficiency

The pricing-efficiency tests estimate:

$$Efficiency_{i,t}^{-1} = \alpha_i + \lambda_t + \beta \text{PostEDGAR}_{i,t} + \gamma' X_{i,t} + \epsilon_{i,t}, \quad (6)$$

where inverse efficiency measures include price synchronicity, absolute return autocorrelation, and pricing error.

Interpretation: A negative β indicates that EDGAR improves price efficiency.

4 Identification and empirical design

4.1 Core identification

- The SEC assigned firms to ten EDGAR phase-in groups.
- Firms in earlier groups become treated while later groups remain untreated.
- Firm fixed effects absorb time-invariant firm differences.
- Time fixed effects absorb aggregate market conditions and trends.
- The estimates ask whether information production changes when a firm becomes an EDGAR filer relative to not-yet-treated firms.

Interpretation: The identifying assumption is that, absent EDGAR, information-production trends for early and late EDGAR cohorts would have evolved similarly conditional on controls and fixed effects.

4.2 Why the design is useful

- EDGAR adoption is not chosen by the firm in the main mandatory phase-in.
- The shock changes information dissemination technology rather than firm fundamentals directly.
- The outcome variables measure outsider information production rather than just direct public-information dissemination.

Interpretation: This allows the paper to separate the effect of better dissemination from ordinary firm-performance or market-wide shocks.

4.3 Addressing alternative explanations

- Propensity-score matching compares treated firms to similar not-yet-treated firms.
- Group-specific time trends address the possibility that EDGAR groups follow different trends.
- Falsification tests define pseudo events before actual EDGAR implementation and find no comparable effect.
- Ease-of-access tests distinguish firms whose filings were initially only partly accessible on the internet.
- Excluding transitional filers removes firms in the early test group CF-01.

Interpretation: The robustness tests make it harder for cohort trends, firm composition, or preexisting changes to explain the results.

4.4 Limitations

- The paper observes trades and analyst actions, not the exact private signals investors and analysts produce.
- Some EDGAR phase-in assignments may correlate with firm characteristics, although controls and matching address this concern.
- The individual-investor sample comes from a brokerage dataset, not the entire population of investors.
- The EDGAR shock is an early-internet setting; later information technologies may have different effects.

Interpretation: The paper's strongest evidence is the consistent alignment of individual trading informativeness, analyst research outcomes, and pricing efficiency.

5 Tables and figures

Visuals are directly cropped from the paper and grouped compactly.

5.1 Figure 1: EDGAR phase-in schedule

Read:

- Figure 1 plots the cumulative number of firms subject to mandatory filing through EDGAR from January 1993 through December 1996.
- The implementation is staggered, with discrete increases as groups switch to mandatory electronic filing.
- The final group begins mandatory filing in May 1996.

Economic message: The stepwise rollout supplies the paper's identifying variation.

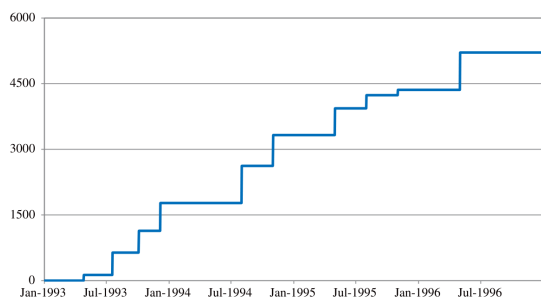


Figure 1
Staggered implementation of mandatory filing through EDGAR

This figure plots the number of firms subject to mandatory filing through EDGAR during the period from January 1993 through December 1996.

Table 1
Summary statistics

A. Summary statistics for the individual trading sample

	# of obs	Mean	SD	P10	Median	P90
Main variables						
<i>Post-EDGAR</i>	29,364	0.431	0.495	0.000	0.000	1.000
<i>Netbuy</i> _[+1,+20] (%)	29,364	0.034	3.560	-1.700	-0.019	1.773
<i>Netbuy by internet users</i> _[+1,+20] (%)	29,364	0.008	1.388	-0.202	0.000	0.200
<i>Netbuy by nonusers</i> _[+1,+20] (%)	29,364	0.026	3.115	-1.424	0.000	1.459
<i>Netbuy by high-RPI investors</i> _[+1,+20] (%)	29,364	-0.003	1.605	-0.452	0.000	0.456
<i>Netbuy by low-RPI investors</i> _[+1,+20] (%)	29,364	0.004	2.032	-0.589	0.000	0.570
<i>CAR</i> _[+21,+83] (%)	29,364	0.591	21.670	-21.542	-0.071	22.778
<i>CAR</i> _[+21,+146] (%)	29,364	1.183	32.064	-29.927	-0.056	32.327
<i>CAR</i> _[+21,+272] (%)	29,364	2.872	49.613	-41.652	0.649	48.226
<i>CAR</i> _[+21,+388] (%)	29,364	4.309	62.065	-50.456	1.557	60.308
Control variables						
<i>SUE decile rank</i>	29,364	4.556	2.874	1.000	5.000	8.000
<i>Total assets</i> (\$ mil)	29,364	2,532.050	7,039.280	24.647	258.121	5,585.520
<i>Book-to-market</i>	29,364	0.652	0.483	0.186	0.548	1.232
<i>Firm age</i> (years)	29,364	17.932	15.675	4.000	13.000	35.000
<i>Prior stock return</i>	29,364	0.284	0.609	-0.301	0.158	1.000
<i>ROA</i>	29,364	0.027	0.127	-0.062	0.043	0.130
<i>Book leverage</i>	29,364	0.495	0.231	0.186	0.490	0.833
<i>Asset tangibility</i>	29,364	0.295	0.230	0.034	0.239	0.667
<i>Sales growth</i>	29,364	0.175	0.376	-0.093	0.100	0.487
<i>CapEx</i>	29,364	0.077	0.084	0.005	0.054	0.162
<i>R&D</i>	29,364	0.046	0.090	0.000	0.000	0.154
<i>Institutional ownership</i>	29,364	0.395	0.221	0.090	0.392	0.694

B. Summary statistics for the analyst sample

	# of obs	Mean	SD	P10	Median	P90
Main variables						
<i>Post-EDGAR</i>	103,929	0.512	0.500	0.000	1.000	1.000
<i># of analysts</i>	103,929	2.489	3.922	0.000	1.000	8.000
<i>Forecast accuracy</i> (x100)	56,447	-0.915	7.943	-1.688	-0.199	0.000
<i>Revision</i> (%)	358,443	-0.184	0.715	-0.642	-0.042	0.193
<i>CAR</i> _[-1,+1] (%)	358,443	-0.232	4.999	-4.790	-0.146	4.549
Control variables						
<i>Total assets</i> (\$ mil)	103,929	2,368.320	12,143.340	12.676	152.707	3,630.230
<i>Book-to-market</i>	103,929	0.723	23.129	0.160	0.576	1.368
<i>Firm age</i> (years)	103,929	16.725	14.805	3.000	12.000	33.000
<i>Prior stock return</i>	103,929	0.240	0.777	-0.364	0.125	0.857
<i>ROA</i>	103,929	-0.001	0.191	-0.123	0.032	0.115
<i>Book leverage</i>	103,929	0.539	0.308	0.193	0.517	0.909
<i>Asset tangibility</i>	103,929	0.292	0.239	0.024	0.231	0.678
<i>Sales growth</i>	103,929	0.356	7.398	-0.124	0.087	0.472
<i>CapEx</i>	103,929	0.083	0.943	0.002	0.048	0.158
<i>R&D</i>	103,929	0.045	0.273	0.000	0.000	0.133
<i>Institutional ownership</i>	103,929	0.319	0.236	0.030	0.285	0.660

This table reports the summary statistics for the individual trading sample (panel A) and analyst sample (panel B). *Post-EDGAR* is an indicator that equals one after a firm-quarter becomes a mandatory EDGAR filer and zero otherwise. *Netbuy*_[+1,+20] is the number of shares bought minus the number of shares sold by individual investors as a fraction of the number of shares outstanding during a 20-day window after an earnings announcement. *Netbuy by internet users*_[+1,+20] and *Netbuy by nonusers*_[+1,+20] are similarly defined for individual investors with access to the internet and those without, respectively. We identify an investor as an internet user if she placed a trade through the internet in the past. *Netbuy by high-RPI investors*_[+1,+20] and *Netbuy by low-RPI investors*_[+1,+20] are defined in a similar manner. We adapt the reliance on public information (RPI) measure proposed by Kacperczyk and Seru (2007) to classify individual investors into two groups based on the median RPI. *CAR*_[+21,+83], *CAR*_[+21,+146], *CAR*_[+21,+272], and *CAR*_[+21,+388] are cumulative DGTW-characteristics-adjusted returns during a 3-, 6-, 12-, and 18-month window starting from the 21st day after an earnings announcement, respectively. *# of analysts* is the number of analysts making quarterly earnings forecasts in the I/B/E/S database for a stock in a given quarter. *Forecast accuracy* is negative of the absolute value of the difference between the actual earnings per share and the median analyst forecast normalized by stock price (following Lang, Lins, and Miller 2003). *Revision* is the difference between two consecutive quarterly earnings forecasts of an analyst for the same stock-quarter scaled by stock price (following Clement and Tse 2003). *CAR*_[-1,+1] is the cumulative DGTW characteristics-adjusted returns during a 3-day window around an earnings forecast revision by an analyst. *SUE decile rank* is the decile rank of earnings surprises, defined as the difference in EPS before extraordinary items between the current quarter and the same quarter of the previous year normalized by stock price (following Jeadeesh and Livnat 2006). The rankings are from zero to nine, with zero (nine) representing

5.2 Table 1: Summary statistics

Read:

- Panel A summarizes the individual trading sample. Mean post-announcement net buying is 0.034%, and the standard deviation is 3.560%.
- The mean 3-month future DGTW-adjusted return is 0.591%, with a standard deviation of 21.670%.

- Panel B summarizes the analyst sample. Mean analyst coverage is 2.489 analysts, and the mean forecast-accuracy measure is -0.009.
- Revision CARs have mean -0.232%.

Economic message: The paper studies a large matched sample of firm-quarters, household trades, and analyst forecasts around earnings announcements.

5.3 Table 2: EDGAR and informativeness of individual investors' trades

Table 2
Staggered implementation of EDGAR and the informativeness of trades by individual investors

A. Baseline specifications

Dependent =	CAR _[+21,+83] (1)	CAR _[+21,+146] (2)	CAR _[+21,+272] (3)	CAR _[+21,+398] (4)
Netbuy _[+1,+20] × Post-EDGAR	0.306 (3.29)***	0.451 (2.53)**	0.406 (2.11)**	0.387 (1.89)*
Netbuy _[+1,+20]	0.019 (0.47)	0.066 (1.23)	0.102 (1.18)	0.229 (2.36)**
Post-EDGAR	-0.019 (2.06)*	-0.036 (3.35)***	-0.086 (5.52)***	-0.096 (5.53)***
Firm FEs	Yes	Yes	Yes	Yes
Year-quarter FEs	Yes	Yes	Yes	Yes
# of observations	29,058	29,058	29,058	29,058
Adj. R-squared	.05	.13	.27	.38

B. Controlling for firm characteristics and their interaction terms with Netbuy

Dependent =	CAR _[+21,+83] (1)	CAR _[+21,+146] (2)	CAR _[+21,+272] (3)	CAR _[+21,+398] (4)
Netbuy _[+1,+20] × Post-EDGAR	0.317 (3.43)***	0.473 (2.68)**	0.459 (2.83)***	0.437 (2.19)**
Netbuy _[+1,+20]	0.312 (1.58)	0.649 (1.98)*	0.389 (0.72)	0.805 (1.47)
Post-EDGAR	-0.016 (1.90)*	-0.028 (2.70)**	-0.074 (5.51)***	-0.080 (5.33)***
Firm controls	Yes	Yes	Yes	Yes
Netbuy _[+1,+20] × Firm controls	Yes	Yes	Yes	Yes
Firm FEs	Yes	Yes	Yes	Yes
Year-quarter FEs	Yes	Yes	Yes	Yes
# of observations	29,058	29,058	29,058	29,058
Adj. R-squared	.10	.19	.36	.47

C. Internet users versus nonusers

Dependent =	CAR _[+21,+83] (1)	CAR _[+21,+146] (2)	CAR _[+21,+272] (3)	CAR _[+21,+398] (4)
Netbuy by internet users × Post-EDGAR (b_1)	0.492 (1.86)*	0.883 (2.34)**	1.145 (3.02)***	1.323 (2.83)***
Netbuy by nonusers × Post-EDGAR (b_2)	0.248 (2.41)**	0.269 (1.27)	0.140 (0.53)	0.064 (0.21)
Netbuy by internet users	0.020 (0.17)	0.140 (0.83)	0.173 (0.85)	0.182 (0.78)
Netbuy by nonusers	0.024 (0.56)	0.063 (0.94)	0.097 (0.82)	0.251 (1.95)*
Post-EDGAR	-0.019 (2.38)**	-0.036 (3.56)***	-0.086 (5.90)***	-0.096 (5.77)***
Firm FEs	Yes	Yes	Yes	Yes
Year-quarter FEs	Yes	Yes	Yes	Yes
p-value for $b_1 = b_2$	[.405]	[.201]	[.045]**	[.040]**
# of observations	29,058	29,058	29,058	29,058
Adj. R-squared	.05	.13	.27	.38

(Continued)

Table 2
(Continued)

D. High- versus low-RPI investors

Dependent =	CAR _[+21,+83] (1)	CAR _[+21,+146] (2)	CAR _[+21,+272] (3)	CAR _[+21,+398] (4)
Netbuy by high-RPI investors × Post-EDGAR (b_1)	-0.033 (0.18)	-0.080 (0.25)	0.040 (0.13)	-0.081 (0.19)
Netbuy by low-RPI investors × Post-EDGAR (b_2)	0.583 (3.11)***	0.982 (3.28)***	0.874 (2.10)**	0.807 (1.63)
Netbuy by high-RPI investors	0.102 (0.95)	0.154 (1.03)	0.186 (1.06)	0.347 (1.73)*
Netbuy by low-RPI investors	-0.032 (0.38)	-0.005 (0.03)	-0.076 (0.34)	0.154 (0.58)
Post-EDGAR	-0.019 (2.37)**	-0.035 (3.53)***	-0.086 (5.86)***	-0.096 (5.72)***
Firm FEs	Yes	Yes	Yes	Yes
Year-quarter FEs	Yes	Yes	Yes	Yes
p-value for $b_1 = b_2$	[.022]**	[.010]**	[.115]	[.231]
# of observations	29,058	29,058	29,058	29,058
Adj. R-squared	.05	.13	.27	.38

This table reports regression analysis of the impact of EDGAR on the informativeness of individual investors' trades about subsequent stock returns. The dependent variables are cumulative DGTW-characteristics-adjusted

Read:

- In baseline specifications, Netbuy × PostEDGAR is positive and significant across all four future-return horizons.
- In panel A, the coefficient is 0.306 for 3-month returns and 0.451 for 6-month returns.
- Adding firm controls leaves the results positive and significant.
- Internet users drive the effect: the internet-user interaction is positive and significant, while the nonuser interaction is weaker.
- Low-RPI investors account for most of the information-production effect; high-RPI investors do not.

Economic message: EDGAR makes individual trades more predictive of future returns, especially for investors with internet access and investors less reliant on raw public signals.

5.4 Table 3: Cross-sectional tests

Read:

- Panel A shows the EDGAR effect is stronger for opaque firms.
- The triple interaction with *Opaque* is positive and significant across all future-return horizons.
- Panel B shows the effect is weaker when institutional presence is high.
- The triple interaction with high institutional presence is negative and significant at shorter horizons.

Economic message: EDGAR matters most where prior information asymmetry is high and where institutions are less likely to have already produced information.

Table 3
Staggered implementation of EDGAR and the informativeness of trades by individual investors:
Cross-sectional tests

A. High versus low information asymmetry stocks

Dependent =	CAR _[+21,+83] (1)	CAR _[+21,+146] (2)	CAR _[+21,+272] (3)	CAR _[+21,+398] (4)
<i>Netbuy</i> _[+1,+20] × <i>Post-EDGAR</i> × <i>Opaque</i>	0.465 (2.25)**	1.110 (3.62)***	0.806 (2.09)**	0.855 (2.17)**
<i>Netbuy</i> _[+1,+20] × <i>Post-EDGAR</i>	0.027 (0.22)	-0.222 (1.11)	-0.107 (0.37)	-0.169 (0.50)
<i>Netbuy</i> _[+1,+20] × <i>Opaque</i>	-0.092 (0.82)	-0.479 (3.62)***	-0.479 (2.60)**	-0.571 (2.52)**
<i>Post-EDGAR</i> × <i>Opaque</i>	-0.005 (0.48)	0.002 (0.15)	-0.001 (0.03)	0.011 (0.35)
<i>Netbuy</i> _[+1,+20]	0.079 (1.09)	0.366 (4.62)***	0.407 (3.71)***	0.594 (4.04)***
<i>Post-EDGAR</i>	-0.017 (1.96)*	-0.034 (3.18)***	-0.084 (5.35)***	-0.095 (5.31)***
<i>Opaque</i>	0.079 (8.89)***	0.133 (11.37)***	0.203 (12.91)***	0.247 (10.31)***
Firm FEs	Yes	Yes	Yes	Yes
Year-quarter FEs	Yes	Yes	Yes	Yes
# of observations	29,058	29,058	29,058	29,058
Adj. R-squared	.06	.14	.29	.39

B. High versus low institutional presence

Dependent =	CAR _[+21,+83] (1)	CAR _[+21,+146] (2)	CAR _[+21,+272] (3)	CAR _[+21,+398] (4)
<i>Netbuy</i> _[+1,+20] × <i>Post-EDGAR</i> × <i>High IP</i>	-0.568 (2.91)***	-0.864 (3.20)***	-0.569 (1.53)	-0.391 (0.93)
<i>Netbuy</i> _[+1,+20] × <i>Post-EDGAR</i>	0.429 (4.63)***	0.636 (2.85)***	0.514 (2.21)**	0.446 (2.06)*
<i>Netbuy</i> _[+1,+20] × <i>High IP</i>	0.167 (1.45)	0.409 (4.06)***	0.483 (3.20)***	0.460 (2.04)*
<i>Post-EDGAR</i> × <i>High IP</i>	0.002 (0.27)	-0.002 (0.16)	0.005 (0.30)	0.007 (0.32)
<i>Netbuy</i> _[+1,+20]	-0.013 (0.29)	-0.017 (0.27)	0.010 (0.09)	0.149 (1.24)
<i>Post-EDGAR</i>	-0.020 (1.96)*	-0.033 (2.99)***	-0.088 (4.53)***	-0.098 (4.46)***
<i>High IP</i>	-0.068 (12.29)***	-0.115 (12.08)***	-0.194 (13.14)***	-0.262 (12.14)***
Firm FEs	Yes	Yes	Yes	Yes
Year-quarter FEs	Yes	Yes	Yes	Yes
# of observations	29,058	29,058	29,058	29,058
Adj. R-squared	.06	.14	.29	.40

This table reports regression analysis of how the impact of EDGAR on the informativeness of individual investors' trades varies with the level of information asymmetry and the presence of institutional investors. The dependent variables are cumulative DGTW-characteristics adjusted returns during a 3-, 6-, 12-, or 18-month window starting from the 21st day after an earnings announcement. Panel A interacts the main variables with *Opaque*, which is an indicator that equals one if the firm has no analyst coverage and below-the-median market capitalization and zero otherwise. Panel B interacts the main variables with an indicator for high institutional presence (*High*

Table 4
Staggered implementation of EDGAR and sell-side analyst research

Dependent =	# of analysts		Forecast accuracy	
	(1)	(2)	(3)	(4)
<i>Post-EDGAR</i>	0.223 (3.07)***	0.230 (3.27)***	0.138 (2.81)**	0.128 (3.35)***
<i>log(Total assets)</i>		0.853 (13.64)***		0.227 (3.55)***
<i>Book-to-market</i>		-0.000 (2.65)**		-0.004 (1.37)
<i>log(Firm age)</i>		-0.032 (0.21)		-0.365 (3.45)***
<i>Prior stock return</i>		-0.078 (2.49)**		0.686 (13.87)***
<i>ROA</i>		0.131 (1.76)*		1.265 (5.55)***
<i>Book leverage</i>		-0.340 (4.35)***		-0.364 (2.60)**
<i>Asset tangibility</i>		0.732 (4.01)***		-0.171 (0.67)
<i>Sales growth</i>		0.000 (1.24)		0.000 (0.67)
<i>CapEx</i>		0.005 (1.37)		0.078 (2.28)**
<i>R&D</i>		0.001 (0.06)		0.464 (2.33)**
<i>Institutional ownership</i>		1.812 (10.44)***		1.072 (6.70)***
Firm FEs	Yes	Yes	Yes	Yes
Year-quarter FEs	Yes	Yes	Yes	Yes
# of observations	103,866	103,866	56,281	56,281
Adj. R-squared	.81	.81	.30	.32

This table reports regression analysis of the impact of EDGAR on analyst coverage and analysts' forecast accuracy. The dependent variable in the first (last) two columns is the number of analysts (forecast accuracy). Table 1 defines all variables. Numbers in parentheses are *t*-statistics based on standard errors two-way clustered by firm and year-quarter. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

5.5 Table 4: EDGAR and sell-side analyst research

Read:

- Post-EDGAR has a positive and significant effect on the number of analysts.
- The coefficient is about 0.223 without firm controls and 0.230 with controls.
- Post-EDGAR also improves forecast accuracy, with coefficients around 0.138 and 0.128.

Economic message: Electronic dissemination does not make analysts redundant. It increases analyst coverage and forecast quality.

5.6 Table 5: EDGAR and market responses to analyst forecast revisions

Read:

- The interaction $Revision \times PostEDGAR$ is positive and significant in all specifications.
- Coefficients are around 0.58 to 0.66.
- The strongest specification includes firm-by-year-quarter and analyst-by-year-quarter fixed effects.

Economic message: Analyst revisions become more informative after EDGAR, suggesting that analysts produce more value-relevant research rather than merely disseminating already public filings.

Table 5
Staggered implementation of EDGAR and market responses to analysts' forecast revisions

Dependent =	Revision CAR _[−1,+1]			
	(1)	(2)	(3)	(4)
$Revision \times Post-EDGAR$	0.641 (4.51)***	0.629 (4.57)***	0.655 (4.65)***	0.582 (4.23)***
$Revision$	0.368 (4.80)***	0.221 (3.05)***	0.202 (2.63)**	0.255 (3.21)***
$Post-EDGAR$	0.001 (1.13)	0.002 (2.41)**	0.002 (2.62)**	
$\log(Total\ assets)$		−0.003 (3.19)***	−0.002 (1.93)*	
$Book-to-market$		0.004 (3.87)***	0.004 (3.65)***	
$\log(Firm\ age)$		−0.004 (2.27)**	−0.004 (2.28)**	
$Prior\ stock\ return$		0.010 (11.85)***	0.011 (11.32)***	
ROA		−0.009 (2.03)*	−0.009 (1.81)*	
$Book\ leverage$		−0.001 (0.31)	0.000 (0.01)	
$Asset\ tangibility$		−0.007 (1.60)	−0.008 (2.00)*	
$Sales\ growth$		−0.000 (0.33)	−0.000 (0.67)	
$CapEx$		−0.001 (0.23)	−0.002 (0.82)	
$R\&D$		0.003 (0.63)	0.002 (0.38)	
$Institutional\ ownership$		0.005 (1.53)	0.004 (1.42)	
Firm FEs	Yes	Yes	Yes	No
Analyst FEs	Yes	Yes	No	No
Year-quarter FEs	Yes	Yes	No	No
Firm $\times$ year-quarter FEs	No	No	No	Yes
Analyst $\times$ year-quarter FEs	No	No	Yes	Yes
# of observations	358,080	354,853	349,114	342,824
Adj. R-squared	.06	.07	.07	.10

This table reports regression analysis of the impact of EDGAR on the informativeness of analysts' forecast revisions. The dependent variable is the cumulative DGTW characteristics-adjusted returns during a 3-day window around analyst revisions. The unit of observation is a revision event. Table 1 defines all variables. Numbers in parentheses are t -statistics based on standard errors three-way clustered by firm, analyst, and year-quarter. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

Table 6
Propensity score matching

A. Pre- and post-matching regressions

Dependent =	Treated	
	Pre-matching (1)	Post-matching (2)
$\log(Total\ assets)$	−0.035 (4.08)***	−0.021 (1.31)
$Book-to-market$	0.149 (5.72)***	−0.043 (1.36)
$\log(Firm\ age)$	0.348 (24.76)***	0.042 (1.34)
$Prior\ stock\ return$	0.133 (1.27)	0.060 (0.47)
ROA	0.053 (1.65)*	−0.036 (1.11)
$Book\ leverage$	0.315 (5.57)***	−0.009 (0.11)
$Asset\ tangibility$	0.002 (0.03)	0.154 (1.26)
$Sales\ growth$	−0.520 (2.46)**	−0.129 (0.52)
$CapEx$	−0.050 (1.83)*	−0.004 (0.39)
$R\&D$	0.008 (0.04)	−0.167 (0.71)
$Institutional\ ownership$	0.047 (0.67)	0.181 (1.48)
Cohort FEs	Yes	Yes
Industry FEs	Yes	Yes
# of observations	26,097	8,388
Pseudo R-squared	.09	.00

B. Diff-in-diff tests using the propensity-score-matched sample

	Informativeness of individual trades (1)	# of analysts (2)	Forecast accuracy (3)	Market responses to revisions (4)
DiD estimate	0.371 (2.21)**	0.243 (1.92)*	0.097 (2.04)**	0.380 (2.45)**

This table reports propensity-score matching diagnostics and the diff-in-diff tests using the propensity-score matched sample. Panel A compares firm characteristics across treatment firms (i.e., firms that switch from a nonfiler to an EDGAR filer in a month) and control firms (i.e., firms remain as a filer or nonfiler in the 12 months before the month under consideration and 12 months after). *Treated* is an indicator that takes the value of one for treatment firms and zero for control firms. Column 1 of Panel A uses the pre-matching sample to estimate a logit regression of *Treated* on firm characteristics. We use the predicted probabilities, or propensity scores, from the estimation and perform one-to-one nearest-neighbor matching with replacement. Column 2 of panel A reports the regression on the propensity score-matched sample. Panel B reports the diff-in-diff tests of the

5.7 Table 6: Propensity-score matching

Read:

- Panel A shows matching balances treated and control firms after propensity-score matching.
- Panel B reports DiD estimates in the matched sample.
- The EDGAR effect remains positive for individual-trade informativeness, analyst coverage, forecast accuracy, and market response to revisions.

Economic message: The main effects are not driven by observable differences between EDGAR cohorts and control firms.

5.8 Table 7: Controlling for group-specific trends

Read:

- Table 7 adds group-specific time trends.

- The individual-trade informativeness coefficient remains positive and significant at 0.359.
- The analyst coverage, forecast accuracy, and revision-response results also remain positive and significant.

Economic message: The results are not explained by broad differential trends across EDGAR phase-in groups.

Table 7
Controlling for group-specific time trends

Dependent =	<i>AbnReturn</i> _[+21,+83] (1)	# of analysts (2)	Forecast accuracy (3)	Revision <i>CAR</i> _[−1,+1] (4)
<i>Netbuy</i> _[+1,+20] × <i>Post-EDGAR</i>	0.359 (3.70)***			
<i>Netbuy</i> _[+1,+20]	0.142 (1.29)			
<i>Post-EDGAR</i>	−0.019 (2.00)**	0.216 (2.94)***	0.143 (3.20)***	
<i>Revision</i> × <i>Post-EDGAR</i>				0.613 (4.66)***
<i>Revision</i>				0.291 (1.73)*
Group × time trends	Yes	Yes	Yes	No
Netbuy × group × time trends	Yes	n/a	n/a	n/a
Revision × group × time trends	n/a	n/a	n/a	Yes
Firm FEs	Yes	Yes	Yes	No
Year-quarter FEs	Yes	Yes	Yes	No
Firm × year-quarter FEs	No	No	No	Yes
Analyst × year-quarter FEs	n/a	n/a	n/a	Yes
Clustering SEs	Firm, year-quarter	Firm, year-quarter	Firm, year-quarter	Firm, analyst, year-quarter
# of observations	29,058	103,866	56,281	342,824
Adj. <i>R</i> -squared	.05	.81	.30	.10

This table reports regression analysis of the impact of EDGAR on various information production measures after adding controls for group-specific time trends. The dependent variables are cumulative DGTW-characteristics adjusted returns during a 3-month window starting from the 21st day after an earnings announcement, the number of analysts, forecast accuracy, and the cumulative DGTW-characteristics-adjusted returns during a 3-day window around analyst revisions, respectively. Table 1 defines all variables. Numbers in parentheses in the first three columns are *t*-statistics based on standard errors two-way clustered by firm and year-quarter, and those in the last column are three-way clustered by firm, analyst, and year-quarter. n/a means not applicable. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

Table 8
Falsification tests

Dependent =	<i>AbnReturn</i> _[+21,+83] (1)	# of analysts (2)	Forecast accuracy (3)	Revision <i>CAR</i> _[−1,+1] (4)
<i>Netbuy</i> _[+1,+20] × “ <i>Post-EDGAR</i> ”	−0.050 (0.56)			
<i>Netbuy</i> _[+1,+20]	0.106 (1.65)			
“ <i>Post-EDGAR</i> ”	0.006 (0.63)	0.085 (0.62)	0.026 (0.47)	
<i>Revision</i> × “ <i>Post-EDGAR</i> ”				0.146 (1.47)
<i>Revision</i>				0.059 (1.11)
Firm FEs	Yes	Yes	Yes	No
Year-quarter FEs	Yes	Yes	Yes	No
Firm × year-quarter FEs	No	No	No	Yes
Analyst × year-quarter FEs	n/a	n/a	n/a	Yes
Clustering SEs	Firm, year-quarter	Firm, year-quarter	Firm, year-quarter	Firm, analyst, year-quarter
# of observations	18,497	67,828	30,983	122,755
Adj. <i>R</i> -squared	.06	.83	.39	.07

This table reports regression analysis of information production activities using a 4-year period preceding the actual EDGAR implementation. We define pseudo events as occurring 24 months prior to the actual implementation. The dependent variables are cumulative DGTW-characteristics-adjusted returns during a 3-month window starting from the 21st day after an earnings announcement, the number of analysts, forecast accuracy, and the cumulative DGTW-characteristics-adjusted returns during a 3-day window around analyst revisions, respectively. “*Post-EDGAR*” is an indicator that equals one for firm-quarters that are in the 2-year window after the pseudo event date and zero for firm-quarters that are in the 2-year window immediately before the pseudo event date. Table 1 defines all other variables. Numbers in parentheses in the first three columns are

5.9 Table 8: Falsification tests

Read:

- Pseudo EDGAR events defined before actual implementation have coefficients that are close to zero and statistically insignificant.
- The falsification tests cover individual-trade informativeness, analyst coverage, forecast accuracy, and revision responses.

Economic message: The absence of pre-effects supports the parallel-trends assumption.

5.10 Table 9: Ease of access to EDGAR filings

Read:

- The paper splits the first four EDGAR groups into an interim period when access was less direct and a later post-EDGAR period.
- The post-EDGAR coefficients remain positive and significant.
- The interim effects are generally smaller or insignificant.

Economic message: The effect is tied to effective internet access to filings, not simply to the formal act of electronic submission.

5.11 Table 10: Excluding group CF-01 firms

Read:

- The results survive excluding Group CF-01, which consists mostly of transitional filers.

Table 9
Ease of access to EDGAR filings

Dependent =	<i>AbnReturn</i> _[+21,+83] (1)	# of analysts (2)	Forecast accuracy (3)	Revision <i>CAR</i> _[−1,+1] (4)
<i>Netbuy</i> _[+1,+20] × <i>Post-EDGAR</i>	0.307 (3.28)***			
<i>Netbuy</i> _[+1,+20] × <i>Interim</i>	0.267 (0.82)			
<i>Netbuy</i> _[+1,+20]	0.019 (0.48)			
<i>Post-EDGAR</i>	−0.031 (2.73)***	0.208 (2.89)***	0.178 (3.09)***	
<i>Interim</i>	0.010 (1.12)	0.282 (1.43)	0.020 (0.42)	
<i>Revision</i> × <i>Post-EDGAR</i>				0.619 (4.37)***
<i>Revision</i> × <i>Interim</i>				0.088 (1.29)
<i>Revision</i>				0.255 (3.21)***
Firm FEs	Yes	Yes	Yes	No
Quarter FEs	Yes	Yes	Yes	No
Firm × year-quarter FEs	No	No	No	Yes
Analyst × year-quarter FEs	n/a	n/a	n/a	Yes
Clustering SEs	Firm, year-quarter	Firm, year-quarter	Firm, year-quarter	Firm, analyst, year-quarter
# of observations	29,058	103,866	56,281	342,824
Adj. <i>R</i> -squared	.05	.81	.30	.10

This table reports regression analysis of the impact of EDGAR on various information production measures when we partition the post-EDGAR period for the first four groups into two periods based on the ease of access to EDGAR filings. Specifically, we redefine the *Post-EDGAR* indicator for the first four groups to take the value of one if the firm-quarter is after January 17, 1994 (when the filings became available to internet users without additional charges) and zero otherwise. *Interim* is an indicator variable that takes the value of one if the firm-quarter falls in the interim period, that is, the time from the starting date of mandated electronic filing to EDGAR to January 16, 1994, for the first four groups of companies and zero otherwise. The dependent variables are cumulative DGTW-characteristics adjusted returns during a 3-month window starting from the 21st day after an earnings announcement, the number of analysts, forecast accuracy, and the cumulative DGTW-characteristics-adjusted returns during a 3-day window around analyst revisions, respectively. Table 1 defines all other variables.

Table 10
Excluding group CF-01 firms

Dependent =	<i>AbnReturn</i> _[+21,+83] (1)	# of analysts (2)	Forecast accuracy (3)	Revision <i>CAR</i> _[−1,+1] (4)
<i>Netbuy</i> _[+1,+20] × <i>Post-EDGAR</i>	0.294 (3.09)***			
<i>Netbuy</i> _[+1,+20]	0.020 (0.52)			
<i>Post-EDGAR</i>	−0.021 (2.12)**	0.237 (2.85)***	0.155 (3.38)***	
<i>Revision</i> × <i>Post-EDGAR</i>				0.567 (4.03)***
<i>Revision</i>				0.261 (3.18)***
Firm FEs	Yes	Yes	Yes	No
Year-quarter FEs	Yes	Yes	Yes	No
Firm × year-quarter FEs	No	No	No	Yes
Analyst × year-quarter FEs	n/a	n/a	n/a	Yes
Clustering SEs	Firm, year-quarter	Firm, year-quarter	Firm, year-quarter	Firm, analyst, year-quarter
# of observations	27,695	100,674	53,514	314,634
Adj. <i>R</i> -squared	.05	.80	.30	.10

This table reports regression analysis of the impact of EDGAR on various information production measures after excluding firms assigned to Group CF-01 (which consists mostly of transitional filers) on the phase-in schedule. The dependent variables are cumulative DGTW-characteristics-adjusted returns during a 3-month window starting from the 21st day after an earnings announcement, the number of analysts, forecast accuracy, and the cumulative DGTW-characteristics-adjusted returns during a 3-day window around analyst revisions, respectively. Table 1 defines all variables. Numbers in parentheses in the first three columns are *t*-statistics based on standard errors two-way clustered by firm and year-quarter, and those in the last column are three-way clustered by firm, analyst, and year-quarter. n/a means not applicable. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

significant, suggesting that individual investors trade more actively in a stock

- Individual-trade informativeness, analyst coverage, forecast accuracy, and revision-response effects remain positive and significant.

Economic message: Transitional filers are not driving the main findings.

5.12 Table 11: Trading volume of individual investors

Read:

- Post-EDGAR increases the trading volume of individual investors.
- The coefficient is 0.393 without controls and 0.339 with controls.
- The increase is driven by internet users; the nonuser coefficients are not statistically significant.
- Low-RPI investors account for much of the increase.

Economic message: EDGAR increases both the informativeness and the quantity of individual trading, with the strongest response among investors most likely to use the new technology productively.

5.13 Table 12: Stock pricing efficiency

Read:

- Post-EDGAR lowers price synchronicity, absolute return autocorrelation, and pricing error.
- The coefficients are negative and statistically significant across the main pricing-efficiency proxies.

Economic message: Greater information dissemination improves price efficiency by encouraging information production.

6 Short summary

- The paper studies how EDGAR's staggered rollout affected information production by market participants.
- The EDGAR shock reduced the cost of obtaining corporate filings and broadened public access to disclosures.

Table 11
Staggered implementation of EDGAR and the trading volume of individual investors

A. All individual investors

Dependent =	Trading volume[+1,+20]	
	(1)	(2)
Post-EDGAR	0.393 (2.29)**	0.339 (2.17)**
SUE decile rank		0.004 (0.31)
log(Total assets)		-0.523 (2.54)**
Book-to-market		0.578 (2.63)**
log(Firm age)		-0.051 (0.13)
Prior stock return		0.799 (7.11)***
ROA		1.724 (2.09)**
Book leverage		1.593 (1.90)*
Asset tangibility		-1.227 (1.02)
Sales growth		-0.006 (0.04)
CapEx		1.844 (2.52)**
R&D		0.150 (0.11)
Institutional ownership		-2.592 (4.30)***
Firm FEs	Yes	Yes
Year-quarter FEs	Yes	Yes
# of observations	29,418	29,418
Adj. R-squared	.24	.25

B. Internet users versus nonusers

Dependent =	Trading volume of internet users [+1,+20]		Trading volume of nonusers[+1,+20]	
	(1)	(2)	(3)	(4)
Post-EDGAR	0.172 (2.80)***	0.161 (2.72)***	0.219 (1.51)	0.176 (1.34)
Firm controls	No	Yes	No	Yes
Firm FEs	Yes	Yes	Yes	Yes
Year-quarter FEs	Yes	Yes	Yes	Yes
# of observations	29,418	29,418	29,418	29,418
Adj. R-squared	.13	.14	.22	.23

C. High- versus low-RPI investors

Dependent =	Trading volume of high-RPI investors[+1,+20]		Trading volume of low-RPI investors[+1,+20]	
	(1)	(2)	(3)	(4)
Post-EDGAR	0.082 (1.23)	0.066 (1.03)	0.320 (3.20)***	0.270 (3.06)***
Firm controls	No	Yes	No	Yes
Firm FEs	Yes	Yes	Yes	Yes
Year-quarter FEs	Yes	Yes	Yes	Yes
# of observations	29,418	29,418	29,418	29,418
Adj. R-squared	.13	.13	.16	.17

This table reports regression analysis of the impact of EDGAR on the trading volume of individual investors following earnings announcements. The dependent variable in panel A is individual investors' trading volume measured as the total number of shares traded by our sample of individual investors (purchases plus sales) during the first 20 trading days following an earnings announcement scaled by the number of shares outstanding. Panel B partitions the trading volume variable into trading volume of

Table 12
Staggered implementation of EDGAR and stock pricing efficiency

Dependent =	Price synchronicity		Abs(Stock return autocorrelation)		Pricing error	
	(1)	(2)	(3)	(4)	(5)	(6)
Post-EDGAR	-0.008 (2.31)**	-0.009 (2.61)**	-0.011 (5.76)***	-0.011 (6.08)***	-0.007 (2.36)**	-0.005 (2.20)**
log(Total assets)		0.012 (5.34)***		-0.014 (9.64)***		-0.022 (10.39)***
Book-to-market		-0.013 (9.33)***		0.013 (8.89)***		0.023 (8.83)***
log(Firm age)		-0.014 (3.29)***		-0.009 (2.56)**		-0.006 (1.36)
Prior stock return		0.014 (10.86)***		-0.026 (21.45)***		-0.036 (26.47)***
ROA		-0.001 (0.49)		-0.006 (1.52)		-0.015 (2.41)**
Book leverage		-0.017 (5.75)***		0.015 (3.45)***		0.033 (5.76)***
Asset tangibility		-0.012 (2.18)**		0.007 (1.07)		0.001 (0.06)
Sales growth		0.002 (3.03)***		-0.001 (1.56)		-0.002 (1.77)*
CapEx		0.022 (3.68)***		-0.031 (4.73)***		-0.038 (4.04)***
R&D		0.006 (0.88)		-0.012 (1.21)		-0.022 (1.63)
Institutional ownership		0.054 (7.98)***		-0.035 (6.80)***		-0.096 (13.05)***
Firm FEs	Yes	Yes	Yes	Yes	Yes	Yes
Year-month FEs	Yes	Yes	Yes	Yes	Yes	Yes
# of observations	300,395	300,395	300,370	300,370	158,319	158,319
Adj. R-squared	.43	.43	.17	.18	.39	.43

- Individual investors' post-earnings net buying became more predictive of future returns after firms became EDGAR filers.
- Internet users and low-RPI investors account for much of this effect.
- Analysts increased coverage and produced more accurate forecasts after EDGAR.
- Analyst forecast revisions generated stronger market reactions after EDGAR.
- Propensity-score matching, group-specific trends, falsification tests, access-timing tests, and exclusion of transitional filers support the main results.
- EDGAR improved pricing efficiency as measured by price synchronicity, return autocorrelation, and pricing-error proxies.
- The evidence favors a crowding-in effect: modern information technology increases information production.

7 Conclusion

7.1 Core conclusion

Gao and Huang show that modern information technology can increase information production in financial markets. EDGAR's electronic dissemination of corporate filings makes individual-investor trades more informative, increases analyst coverage and accuracy, and improves pricing efficiency.

7.2 Economic intuition

Better dissemination changes incentives. When filings become easier to access and process, some investors and analysts spend more effort producing information. The strongest effects appear for internet users, low-RPI investors, opaque firms, and firms with weaker institutional information production before EDGAR.

7.3 Takeaway

The paper's main message is that public information technology does not necessarily crowd out private information production. In this setting, lowering information-access costs crowds in outsider research and makes prices more informative.